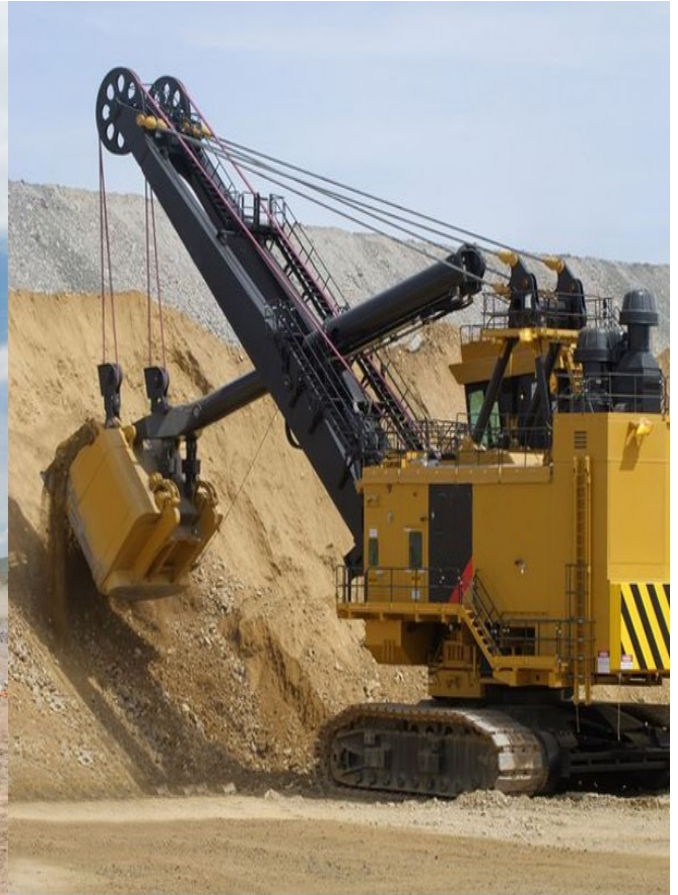


Qualification Pack



Shovel Operator

Electric Rope Shovel/ Hydraulic Shovel

QP Code: MIN/Q1408

Version: 1.0

NSQF Level: 4.0

SCMS (Skill council of Mining Sector) || FIMI House, B-311, Okhla Industrial Area, Phase-I
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Qualification Pack

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MIN/Q1408: Shovel Operator

Brief Job Description

A Shovel Operator is responsible for operating large hydraulic or electric-powered rope mining shovels used in mining operations. The individual plays a crucial role in excavating and loading materials such as ore, overburden, or minerals onto haul trucks or conveyor systems. The person performs appropriate pre-operation checks on the shovel and its basic repair and maintenance.

Personal Attributes

This job requires the individual to concentrate on the job at hand and complete it efficiently and effectively. The individual should be physically agile, strong, have good eye sight and not suffer from colour-blindness.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MIN/N1419: Perform the pre-operation checks on the shovel](#)
2. [MIN/N1420: Operate the shovel for mining operations](#)
3. [MIN/N1421: Perform the post-shovel operation activities](#)
4. [MIN/N1422: Carry out appropriate activities before and after the shovel maintenance](#)
5. [MIN/N1425: Follow the mining plan and instructions for material handling](#)
6. [MIN/N1426: Carry out basic repair and maintenance of the mining shovel](#)
7. [MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines](#)
8. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Electives (mandatory to select at least one):

Elective 1: Electric Rope Shovel

1. [MIN/N1423: Operate and maintain the electric rope shovel](#)

Elective 2: Hydraulic Shovel

1. [MIN/N1424: Operate and maintain the hydraulic shovel](#)

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Qualification Pack (QP) Parameters

Sector	Mining
Sub-Sector	Mining Operation
Occupation	Loading and Hauling - Opencast
Country	India
NSQF Level	4.0
Credits	27
Aligned to NCO/ISCO/ISIC Code	NCO-2015/8111.2700
Minimum Educational Qualification & Experience	10th grade pass (+2-year NTC/NAC or equivalent)(Mechanical/ Mining/Civil/ Electrical) OR Completed 2nd year of the 3-year diploma after 10 (Mechanical/ Mining/Civil/ Electrical) OR 12th grade Pass (or equivalent)(+0.5 year relevant experience) OR 11th grade pass (+1.5 years relevant experience) OR 10th grade pass (or equivalent)(+3 years relevant experience) OR Previous relevant Qualification of NSQF Level (3.0)(+3 years relevant experience)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	Heavy Motor Vehicle Driving License
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	11/03/2029
NSQF Approval Date	11/03/2026
Version	1.0
Reference code on NQR	QG-04-Mining-05068-2026-V1-SCMS
NQR Version	1.0

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Remarks:

Total Duration: 690 Hours = 660 Hours Domain + 30 Hours Employability Skills

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MIN/N1419: Perform the pre-operation checks on the shovel

Description

This unit is about performing the appropriate pre-operation checks on the shovel to ensure its readiness for use in mining operations.

Scope

The scope covers the following :

- Perform routine checks on the shovel
- Check the emergency equipment

Elements and Performance Criteria

Perform the pre-operation checks on the shovel

To be competent, the user/individual on the job must be able to:

- PC1.** check the shovel components for signs of wear and tear or damage, and repair or replace them as appropriate
- PC2.** check for loose components and tighten them appropriately using the recommended tools
- PC3.** check for fluid leaks and take appropriate corrective measures to seal the leaks
- PC4.** examine the safety devices, such as seat belts and emergency shut-off switches for the correct functioning
- PC5.** check the hydraulic fluid, engine oil, coolant, and fuel levels and refill them as required to maintain the recommended levels
- PC6.** inspect the tires or tracks for wear, damage, or punctures, and replace or repair them as appropriate
- PC7.** check the controls and instruments, such as levers and pedals for the correct functioning
- PC8.** follow the manufacturer-provided checklist for performing a thorough safety inspection of the shovel
- PC9.** maintain the record of pre-operation checks

Check the emergency equipment

To be competent, the user/individual on the job must be able to:

- PC10.** test the fire extinguishers for the correct functioning
- PC11.** check the first-aid kit is updated with the necessary items
- PC12.** ensure the emergency equipment is readily accessible to deal with emergencies promptly

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of conducting pre-operation checks on a mining shovel
- KU2.** the major components of the mining shovel, including the boom, dipper, bucket, hydraulic systems, tracks or tires, engine, electrical systems, and controls

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- KU3.** the safety protocols and procedures for conducting pre-operation checks
- KU4.** the potential hazards associated with shovel operation and the importance of mitigating these risks
- KU5.** the importance of following the manufacturer's equipment manuals and pre-operation inspection checklist
- KU6.** the visual inspection of the shovel for damage, wear, or abnormalities
- KU7.** the common signs of equipment issues, such as fluid leaks, loose bolts, damaged hoses, or worn components
- KU8.** the functions of different types of fluids used in the shovel, i.e. hydraulic fluid, engine oil, coolant, fuel
- KU9.** how to check fluid levels, including proper oil and coolant levels, and assess the fluid quality, i.e. cleanliness and colour
- KU10.** the types of undercarriage used in shovels, i.e. tires or tracks
- KU11.** the inspection of tires or tracks for wear, damage, tension, and alignment
- KU12.** the electrical systems used in the shovel, including lights, alarms, and safety devices
- KU13.** the inspection of electrical components for damage or malfunction
- KU14.** the importance of safety features in equipment operation
- KU15.** the location and operation of safety devices, such as seat belts, emergency shut-off switches, fire extinguishers, and backup alarms
- KU16.** the recommended lubrication and greasing points on the shovel
- KU17.** how to apply grease and lubricants to critical components as specified in the maintenance manual
- KU18.** the basic understanding of hydraulic systems and how to identify and assess fluid leaks in them
- KU19.** the implications of fluid leaks on equipment operation and safety
- KU20.** the importance of properly torqued bolts and fasteners in equipment safety and performance
- KU21.** the inspection of the tightness of critical bolts and fasteners
- KU22.** the importance of maintaining accurate records of pre-operation checks, including any issues identified and actions taken
- KU23.** the importance of checking the equipment's maintenance history and any previous issues

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain the appropriate data and records
- GS2.** read the appropriate reports and literature concerning the field of work
- GS3.** communicate professionally with all the stakeholders
- GS4.** listen attentively to understand the information/ instructions being shared and take appropriate action
- GS5.** coordinate with co-workers to achieve the work objectives
- GS6.** plan tasks based on priority to ensure their timely completion



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- GS7.** identify possible disruptions to work and take appropriate mitigation measures
- GS8.** take prompt action to deal with workplace emergencies and accidents
- GS9.** evaluate all possible solutions to work-related problems and select the best one
- GS10.** follow the recommended practices for the timely completion of work and achievement of organizational objectives

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform the pre-operation checks on the shovel</i>	23	38	-	16
PC1. check the shovel components for signs of wear and tear or damage, and repair or replace them as appropriate	3	5	-	2
PC2. check for loose components and tighten them appropriately using the recommended tools	2	4	-	2
PC3. check for fluid leaks and take appropriate corrective measures to seal the leaks	2	4	-	2
PC4. examine the safety devices, such as seat belts and emergency shut-off switches for the correct functioning	3	4	-	2
PC5. check the hydraulic fluid, engine oil, coolant, and fuel levels and refill them as required to maintain the recommended levels	3	4	-	2
PC6. inspect the tires or tracks for wear, damage, or punctures, and replace or repair them as appropriate	3	5	-	2
PC7. check the controls and instruments, such as levers and pedals for the correct functioning	3	4	-	2
PC8. follow the manufacturer-provided checklist for performing a thorough safety inspection of the shovel	2	4	-	1
PC9. maintain the record of pre-operation checks	2	4	-	1
<i>Check the emergency equipment</i>	7	12	-	4
PC10. test the fire extinguishers for the correct functioning	3	4	-	2
PC11. check the first-aid kit is updated with the necessary items	2	4	-	1
PC12. ensure the emergency equipment is readily accessible to deal with emergencies promptly	2	4	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1419
NOS Name	Perform the pre-operation checks on the shovel
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Loading and Hauling - Opencast
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	11/03/2026
Next Review Date	11/03/2029
NSQC Clearance Date	11/03/2026

Qualification Pack

MIN/N1420: Operate the shovel for mining operations

Description

This unit is about operating the shovel during mining operations for effective mining.

Scope

The scope covers the following :

- Prepare for the shovelling operation
- Excavate the material
- Dump the material
- Maintain effective communication with other personnel
- Follow the safety protocols

Elements and Performance Criteria

Prepare for the shoveling operation

To be competent, the user/individual on the job must be able to:

- PC1.** determine the location of the material to be excavated, the haul trucks' routes, and the dumping points
- PC2.** follow the appropriate safety practices to enter and exit the operator's cabin to prevent slips or falls
- PC3.** follow the manufacturer's startup procedure, including turning on the ignition, engaging the parking brake, and starting the engine
- PC4.** ensure appropriate adjustment of the operator's seat and controls for optimal visibility and control
- PC5.** follow the appropriate practices for effective engine performance, such as allowing the engine to warm up for the appropriate duration
- PC6.** use the controls to maneuver the shovel into position for excavation and material handling tasks
- PC7.** ensure the shovel is on stable ground with the surrounding area clear of obstacles and personnel

Excavate the material

To be competent, the user/individual on the job must be able to:

- PC8.** use the controls to operate the shovel's boom, dipper, and bucket (or other attachment) to dig into the material
- PC9.** use the bucket or attachment safely, engaging it steadily to pick up and release material
- PC10.** follow the recommended loading techniques to ensure a full and evenly distributed load
- PC11.** adhere to the shovel's load capacity to maintain its stability and prevent tipping
- PC12.** use the appropriate digging technique according to the type of material being excavated
- PC13.** maintain control of the shovel for its safe operation, avoiding sudden movements
- PC14.** use mirrors and cameras (if available) to be aware of the surroundings and the movement of other equipment and personnel in the mining area

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PC15. use the shovel's control system as per the manufacturer's instructions

Dump the material

To be competent, the user/individual on the job must be able to:

PC16. swing the shovel's boom appropriately to position the loaded bucket over the haul truck or material-handling equipment

PC17. dump the material into the designated area or truck bed safely

PC18. follow safe loading and unloading procedures, ensuring the bucket or attachment is positioned correctly

PC19. follow the appropriate practices to optimize the cycle time, loading trucks efficiently and safely

PC20. adhere to the project timelines and applicable quality standards

Maintain effective communication with other personnel

To be competent, the user/individual on the job must be able to:

PC21. maintain effective communication with other equipment operators, truck drivers, and personnel in the mining area

PC22. use radios or hand signals as appropriate to coordinate movements

Follow the safety protocols

To be competent, the user/individual on the job must be able to:

PC23. maintain the shovel at a safe distance from other equipment, vehicles, workers, and obstacles

PC24. identify potential hazards, such as overhead power lines, unstable ground, or falling rocks, and take appropriate safety measures

PC25. operate the shovel appropriately to maintain its control and prevent any safety incidents

PC26. ensure no unauthorized personnel has access to the shovel to prevent any safety incidents

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the use of appropriate make and models of mining shovels

KU2. the capabilities, limitations, and technical specifications of different types of shovels used in mining operations

KU3. the safety protocols and procedures specific to mining shovel operation

KU4. how to mitigate the potential hazards and risks associated with mining operations, e.g. hazards concerning the use of heavy equipment

KU5. the importance of conducting pre-operation inspection of the shovel, including visual checks for damage, wear, and fluid leaks

KU6. the benefits of following the equipment manufacturer's pre-operation inspection checklist and recommendations

KU7. how to operate and manipulate the controls, joysticks, levers, pedals, and switches in the operator's cabin

KU8. the various instrumentation and displays used for monitoring the shovel's performance and status

KU9. the principles of material handling, including digging, loading, swinging, and dumping

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- KU10.** the use of different digging techniques depending on the type of material being excavated
- KU11.** the shovel's load capacity and the importance of adhering to it
- KU12.** how to distribute loads evenly and safely within the shovel's bucket or attachment
- KU13.** the importance of effective communication with other equipment operators, truck drivers, and personnel in the mining area
- KU14.** the use of radios or hand signals to coordinate movements and ensure safety
- KU15.** the appropriate practices to be followed for effective lubrication and hydraulic system performance
- KU16.** the routine shovel maintenance tasks, such as lubrication and greasing
- KU17.** the signs of equipment issues or abnormal operation that require immediate attention
- KU18.** the environmental regulations and guidelines relevant to the mining site operations
- KU19.** the appropriate practices to be followed to minimize environmental impact during shovel operation
- KU20.** the relevant emergency procedures, including evacuation routes and communication protocols in case of accidents or equipment failure
- KU21.** how to respond quickly and effectively in emergencies
- KU22.** the importance of being aware of ground conditions to prevent equipment from getting stuck or tipping over
- KU23.** the impact of weather conditions, such as rain on ground stability
- KU24.** the importance of following the mining plan and material handling plan
- KU25.** the importance of identifying the location of the material to be excavated, haul truck routes, and dumping points
- KU26.** the relevant regulations, including those related to equipment operation, safety, and environmental protection
- KU27.** the importance of staying updated with advancements in mining equipment technology and safety practices

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain the appropriate data and records
- GS2.** read the appropriate reports and literature concerning the field of work
- GS3.** communicate professionally with all the stakeholders
- GS4.** listen attentively to understand the information/ instructions being shared and take appropriate action
- GS5.** coordinate with co-workers to achieve the work objectives
- GS6.** plan tasks based on priority to ensure their timely completion
- GS7.** identify possible disruptions to work and take appropriate mitigation measures
- GS8.** take prompt action to deal with workplace emergencies and accidents
- GS9.** evaluate all possible solutions to work-related problems and select the best one
- GS10.** follow the recommended practices for the timely completion of work and achievement of organizational objectives

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for the shoveling operation</i>	10	13	-	6
PC1. determine the location of the material to be excavated, the haul trucks' routes, and the dumping points	2	2	-	1
PC2. follow the appropriate safety practices to enter and exit the operator's cabin to prevent slips or falls	2	1	-	1
PC3. follow the manufacturer's startup procedure, including turning on the ignition, engaging the parking brake, and starting the engine	1	2	-	1
PC4. ensure appropriate adjustment of the operator's seat and controls for optimal visibility and control	2	2	-	1
PC5. follow the appropriate practices for effective engine performance, such as allowing the engine to warm up for the appropriate duration	1	2	-	1
PC6. use the controls to maneuver the shovel into position for excavation and material handling tasks	1	2	-	1
PC7. ensure the shovel is on stable ground with the surrounding area clear of obstacles and personnel	1	2	-	-
<i>Excavate the material</i>	8	14	-	6
PC8. use the controls to operate the shovel's boom, dipper, and bucket (or other attachment) to dig into the material	1	1	-	1
PC9. use the bucket or attachment safely, engaging it steadily to pick up and release material	1	1	-	-
PC10. follow the recommended loading techniques to ensure a full and evenly distributed load	1	2	-	1
PC11. adhere to the shovel's load capacity to maintain its stability and prevent tipping	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. use the appropriate digging technique according to the type of material being excavated	1	2	-	1
PC13. maintain control of the shovel for its safe operation, avoiding sudden movements	1	2	-	1
PC14. use mirrors and cameras (if available) to be aware of the surroundings and the movement of other equipment and personnel in the mining area	1	2	-	1
PC15. use the shovel's control system as per the manufacturer's instructions	1	2	-	-
<i>Dump the material</i>	6	10	-	3
PC16. swing the shovel's boom appropriately to position the loaded bucket over the haul truck or material-handling equipment	2	2	-	1
PC17. dump the material into the designated area or truck bed safely	1	2	-	1
PC18. follow safe loading and unloading procedures, ensuring the bucket or attachment is positioned correctly	1	2	-	-
PC19. follow the appropriate practices to optimize the cycle time, loading trucks efficiently and safely	1	2	-	1
PC20. adhere to the project timelines and applicable quality standards	1	2	-	-
<i>Maintain effective communication with other personnel</i>	2	4	-	2
PC21. maintain effective communication with other equipment operators, truck drivers, and personnel in the mining area	1	2	-	1
PC22. use radios or hand signals as appropriate to coordinate movements	1	2	-	1
<i>Follow the safety protocols</i>	4	9	-	3
PC23. maintain the shovel at a safe distance from other equipment, vehicles, workers, and obstacles	1	4	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. identify potential hazards, such as overhead power lines, unstable ground, or falling rocks, and take appropriate safety measures	1	2	-	1
PC25. operate the shovel appropriately to maintain its control and prevent any safety incidents	1	2	-	1
PC26. ensure no unauthorized personnel has access to the shovel to prevent any safety incidents	1	1	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1420
NOS Name	Operate the shovel for mining operations
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Loading and Hauling - Opencast
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	11/03/2026
Next Review Date	11/03/2029
NSQC Clearance Date	11/03/2026

Qualification Pack

MIN/N1421: Perform the post-shovel operation activities

Description

This unit is about performing the appropriate activities after the use of shovel in mining operations.

Scope

The scope covers the following :

- Perform the post-operation activities
- Carry out documentation and reporting

Elements and Performance Criteria

Perform the post-operation activities

To be competent, the user/individual on the job must be able to:

- PC1.** follow the manufacturer's shutdown procedure, i.e. turning off the engine, disengaging controls, and applying the parking brake
- PC2.** ensure the equipment is allowed to cool down after prolonged use
- PC3.** check for and remove any loose debris or materials that may have been displaced during the shovelling process
- PC4.** clean the shovel and any other tools used during the shovelling operation
- PC5.** carry out surface restoration in the mining area, as applicable
- PC6.** inspect the shovel for any damage or wear and tear that may need to be repaired or addressed before the next use
- PC7.** carry out basic maintenance of the shovel before the next use
- PC8.** coordinate the disposal of any waste materials or debris generated during the shovelling operation
- PC9.** coordinate with the supervisor to resolve any issues or challenges that need to be addressed in future shovelling operations
- PC10.** identify and report any issues or abnormalities observed during the shovel operation to the relevant personnel

Carry out documentation and reporting

To be competent, the user/individual on the job must be able to:

- PC11.** maintain appropriate documentation concerning the shovel operation and completion of assigned tasks
- PC12.** carry out basic reporting for project management and tracking purposes

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the appropriate shutdown procedure for the mining shovel

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- KU2.** the appropriate post-operation measures to be taken to ensure shovel efficiency, e.g. inspection and basic shovel maintenance after every use
- KU3.** the appropriate waste management procedures at mining sites
- KU4.** the importance and process of surface restoration in the mining area
- KU5.** the importance of identifying and resolving any issues or challenges in shovelling operations
- KU6.** the signs of any issues or abnormalities in the mining shovel operation
- KU7.** the appropriate documentation and reporting requirements

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain the appropriate data and records
- GS2.** read the appropriate reports and literature concerning the field of work
- GS3.** communicate professionally with all the stakeholders
- GS4.** listen attentively to understand the information/ instructions being shared and take appropriate action
- GS5.** coordinate with co-workers to achieve the work objectives
- GS6.** plan tasks based on priority to ensure their timely completion
- GS7.** identify possible disruptions to work and take appropriate mitigation measures
- GS8.** take prompt action to deal with workplace emergencies and accidents
- GS9.** evaluate all possible solutions to work-related problems and select the best one
- GS10.** follow the recommended practices for the timely completion of work and achievement of organizational objectives

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform the post-operation activities</i>	24	42	-	16
PC1. follow the manufacturer's shutdown procedure, i.e. turning off the engine, disengaging controls, and applying the parking brake	3	4	-	2
PC2. ensure the equipment is allowed to cool down after prolonged use	2	4	-	2
PC3. check for and remove any loose debris or materials that may have been displaced during the shovelling process	3	4	-	1
PC4. clean the shovel and any other tools used during the shovelling operation	2	4	-	1
PC5. carry out surface restoration in the mining area, as applicable	2	4	-	1
PC6. inspect the shovel for any damage or wear and tear that may need to be repaired or addressed before the next use	2	5	-	1
PC7. carry out basic maintenance of the shovel before the next use	2	6	-	2
PC8. coordinate the disposal of any waste materials or debris generated during the shovelling operation	2	4	-	2
PC9. coordinate with the supervisor to resolve any issues or challenges that need to be addressed in future shovelling operations	3	3	-	2
PC10. identify and report any issues or abnormalities observed during the shovel operation to the relevant personnel	3	4	-	2
<i>Carry out documentation and reporting</i>	6	8	-	4
PC11. maintain appropriate documentation concerning the shovel operation and completion of assigned tasks	3	4	-	2



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. carry out basic reporting for project management and tracking purposes	3	4	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1421
NOS Name	Perform the post-shovel operation activities
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Loading and Hauling - Opencast
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	11/03/2026
Next Review Date	11/03/2029
NSQC Clearance Date	11/03/2026

Qualification Pack

MIN/N1422: Carry out appropriate activities before and after the shovel maintenance

Description

This unit is about carrying out appropriate pre and post-shovel maintenance activities.

Scope

The scope covers the following :

- Prepare for shovel maintenance
- Perform the pre-maintenance inspection
- Perform the post-maintenance testing
- Carry out appropriate documentation

Elements and Performance Criteria

Prepare for shovel maintenance

To be competent, the user/individual on the job must be able to:

- PC1.** ensure the shovel is parked on a level surface and is securely isolated from power sources or energy systems
- PC2.** engage all safety locks and mechanisms to prevent accidental operation
- PC3.** check for the availability of appropriate tools, equipment, accessories and spare parts for the maintenance of the shovel

Perform the pre-maintenance inspection

To be competent, the user/individual on the job must be able to:

- PC4.** perform a visual inspection of the entire shovel to identify any issues or areas that need attention
- PC5.** check for any loose bolts, worn-out/damaged components, fluid leaks, or unusual wear

Perform the post-maintenance testing

To be competent, the user/individual on the job must be able to:

- PC6.** carry out post-maintenance testing of the shovel, testing its operations to ensure all the components are functioning correctly
- PC7.** check there are no fluid leaks, unusual noises, or warning indicators

Carry out appropriate documentation

To be competent, the user/individual on the job must be able to:

- PC8.** maintain the records of all maintenance activities, including inspections, repairs, and component replacements
- PC9.** maintain the record of any issues or abnormalities discovered during maintenance and coordinate with the supervisor to resolve them

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** the importance of basic maintenance of a mining shovel to ensure the equipment's reliability, safety, and longevity, prevention of breakdowns, reduction in downtime, equipment effectiveness
- KU2.** the importance of carrying out maintenance of different components of the shovel as per the maintenance schedule recommended by the manufacturer
- KU3.** the importance of using the manufacturer-recommended maintenance tools and spare parts
- KU4.** different models of shovels used in mining operations
- KU5.** the components, systems, and technical specifications of different types of shovels
- KU6.** the safety protocols and procedures specific to maintenance of mining shovels
- KU7.** the potential hazards associated with mining shovel's maintenance and the use of appropriate Personal Protective Equipment (PPE)
- KU8.** how to read and follow the manufacturer's equipment maintenance manual for basic equipment maintenance
- KU9.** the recommended maintenance schedule, procedures, and checklists for a shovel's maintenance
- KU10.** the relevant tools, equipment, and materials required for a shovel's maintenance
- KU11.** the use of different hand and power tools, and specialized equipment
- KU12.** the types of fluids used in the shovel, i.e. hydraulic fluid, engine oil, coolant, and fuel
- KU13.** how to check fluid levels, assess fluid quality, and perform fluid changes
- KU14.** the recommended lubrication points on the shovel and the proper lubricants to be used
- KU15.** how to apply grease and lubricants to critical components to prevent wear and friction
- KU16.** the various filters, i.e. air filters, fuel filters, hydraulic filters, used in a shovel
- KU17.** how to replace or clean filters according to the maintenance schedule
- KU18.** the basic maintenance/troubleshooting of the shovel's electrical systems, including lights, alarms, sensors, and wiring
- KU19.** how to replace the damaged electrical components
- KU20.** the hydraulic systems, hoses, and fittings of the shovel
- KU21.** how to identify and address hydraulic leaks, replace hoses, and perform system checks
- KU22.** the importance of properly torqued bolts and fasteners in equipment safety and performance
- KU23.** the process of visually inspecting and checking the tightness of critical bolts and fasteners
- KU24.** the location and use of safety devices, such as emergency shut-off switches and fire suppression systems
- KU25.** how to test and maintain safety features
- KU26.** the environmental regulations and guidelines relevant to the maintenance of mining shovel
- KU27.** the importance and process of maintaining records of maintenance activities, including tasks performed, dates, and any issues identified
- KU28.** the utilization of maintenance logs and documentation to track equipment history
- KU29.** the emergency response procedures, including the location of emergency equipment and evacuation routes in case of accidents during maintenance

Generic Skills (GS)

Qualification Pack

User/individual on the job needs to know how to:

- GS1.** maintain the appropriate data and records
- GS2.** read the appropriate reports and literature concerning the field of work
- GS3.** communicate professionally with all the stakeholders
- GS4.** listen attentively to understand the information/ instructions being shared and take appropriate action
- GS5.** coordinate with co-workers to achieve the work objectives
- GS6.** plan tasks based on priority to ensure their timely completion
- GS7.** identify possible disruptions to work and take appropriate mitigation measures
- GS8.** take prompt action to deal with workplace emergencies and accidents
- GS9.** evaluate all possible solutions to work-related problems and select the best one
- GS10.** follow the recommended practices for the timely completion of work and achievement of organizational objectives

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for shovel maintenance</i>	11	16	-	7
PC1. ensure the shovel is parked on a level surface and is securely isolated from power sources or energy systems	4	7	-	3
PC2. engage all safety locks and mechanisms to prevent accidental operation	4	4	-	2
PC3. check for the availability of appropriate tools, equipment, accessories and spare parts for the maintenance of the shovel	3	5	-	2
<i>Perform the pre-maintenance inspection</i>	6	9	-	5
PC4. perform a visual inspection of the entire shovel to identify any issues or areas that need attention	3	4	-	2
PC5. check for any loose bolts, worn-out/damaged components, fluid leaks, or unusual wear	3	5	-	3
<i>Perform the post-maintenance testing</i>	7	13	-	4
PC6. carry out post-maintenance testing of the shovel, testing its operations to ensure all the components are functioning correctly	3	6	-	2
PC7. check there are no fluid leaks, unusual noises, or warning indicators	4	7	-	2
<i>Carry out appropriate documentation</i>	6	12	-	4
PC8. maintain the records of all maintenance activities, including inspections, repairs, and component replacements	3	5	-	2
PC9. maintain the record of any issues or abnormalities discovered during maintenance and coordinate with the supervisor to resolve them	3	7	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1422
NOS Name	Carry out appropriate activities before and after the shovel maintenance
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Loading and Hauling - Opencast
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	11/03/2026
Next Review Date	11/03/2029
NSQC Clearance Date	11/03/2026

Qualification Pack

MIN/N1425: Follow the mining plan and instructions for material handling

Description

This unit is about following the mining plan and instructions for material handling.

Scope

The scope covers the following :

- Determine the mining requirements
- Follow the mining plan

Elements and Performance Criteria

Determine the mining requirements

To be competent, the user/individual on the job must be able to:

- PC1.** determine the mining objectives and the planned mining activities by checking the mining plan
- PC2.** check the mining plan to get information regarding the geology, reserves, design, schedule, assigned tasks, etc
- PC3.** determine the applicable legal and regulatory requirements, safety protocols, and environmental standards
- PC4.** identify and determine the roles of other personnel involved in the mining operation
- PC5.** obtain more clarity and plan coordination of mining activities by participating in the relevant meetings and briefings

Follow the mining plan

To be competent, the user/individual on the job must be able to:

- PC6.** perform the assigned tasks and duties according to the mining plan and instructions
- PC7.** use the appropriate tools and equipment as per the mining plan
- PC8.** follow the applicable standard procedures to ensure safety and achieve efficiency
- PC9.** monitor and record own progress and performance as per the organizational procedure
- PC10.** follow the appropriate measures for the protection of the mineral resources and environment
- PC11.** ensure adherence to the mining plan, instructions and applicable safety protocols provided for material handling activities
- PC12.** follow the appropriate methods for the extraction of minerals from the mine
- PC13.** identify and report any problems or deviations from the mining plan
- PC14.** follow the supervisor's instructions, and seek guidance when required
- PC15.** identify and suggest the requirement of appropriate changes in the mining plan based on the relevant new information, and changing conditions or unforeseen circumstances in the mine
- PC16.** identify the scope of improvement in the implementation of the mining plan and share appropriate feedback following the organizational procedure

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** the benefits of the mining plan that outlines the objectives, methods, and procedures for the extraction and processing of minerals from a mine
- KU2.** the role of the mining plan in ensuring the scientific development, conservation, and environmental protection of the mineral resources
- KU3.** the importance for all the mining personnel to understand and follow the mining plan and instructions
- KU4.** the importance of following the mining plan for effective
- KU5.** the importance and methods of effective communication among the mining personnel, supervisors, contractors, etc
- KU6.** the importance of reviewing and updating the mining plan based on any new information, and changing conditions or unforeseen circumstances in the mine
- KU7.** the importance of executing material handling tasks in accordance with the mining plan and instructions provided by supervisors or mine planners
- KU8.** the appropriate measures to be taken to ensure the safe and efficient movement of materials in the mining area
- KU9.** the importance of prioritizing safety by following established safety protocols and procedures throughout the mining operations, e.g. equipment operation, excavation, loading, and unloading of ore/ minerals
- KU10.** the importance of effective communication with supervisors, equipment operators, and other personnel involved in mining and material handling activities
- KU11.** the appropriate records to be maintained to ensure compliance with inventory control procedures, e.g. material quantities, types, destinations, etc
- KU12.** the appropriate mining and material handling to be followed to minimize the environmental impact and ensure compliance with the regulatory requirements
- KU13.** the identification and resolution of common operational issues, such as material spillage, equipment blockages, or handling challenges

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain the appropriate data and records
- GS2.** read the appropriate reports and literature concerning the field of work
- GS3.** communicate professionally with all the stakeholders
- GS4.** listen attentively to understand the information/ instructions being shared and take appropriate action
- GS5.** coordinate with co-workers to achieve the work objectives
- GS6.** plan tasks based on priority to ensure their timely completion
- GS7.** identify possible disruptions to work and take appropriate mitigation measures
- GS8.** take prompt action to deal with workplace emergencies and accidents
- GS9.** evaluate all possible solutions to work-related problems and select the best one

Qualification Pack

GS10. follow the recommended practices for the timely completion of work and achievement of organizational objectives

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Determine the mining requirements</i>	9	15	-	5
PC1. determine the mining objectives and the planned mining activities by checking the mining plan	3	3	-	1
PC2. check the mining plan to get information regarding the geology, reserves, design, schedule, assigned tasks, etc	2	3	-	1
PC3. determine the applicable legal and regulatory requirements, safety protocols, and environmental standards	1	3	-	1
PC4. identify and determine the roles of other personnel involved in the mining operation	1	3	-	1
PC5. obtain more clarity and plan coordination of mining activities by participating in the relevant meetings and briefings	2	3	-	1
<i>Follow the mining plan</i>	21	35	-	15
PC6. perform the assigned tasks and duties according to the mining plan and instructions	2	3	-	2
PC7. use the appropriate tools and equipment as per the mining plan	2	3	-	2
PC8. follow the applicable standard procedures to ensure safety and achieve efficiency	2	3	-	1
PC9. monitor and record own progress and performance as per the organizational procedure	2	3	-	2
PC10. follow the appropriate measures for the protection of the mineral resources and environment	2	3	-	1
PC11. ensure adherence to the mining plan, instructions and applicable safety protocols provided for material handling activities	2	3	-	1
PC12. follow the appropriate methods for the extraction of minerals from the mine	2	4	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. identify and report any problems or deviations from the mining plan	1	3	-	1
PC14. follow the supervisor's instructions, and seek guidance when required	2	3	-	1
PC15. identify and suggest the requirement of appropriate changes in the mining plan based on the relevant new information, and changing conditions or unforeseen circumstances in the mine	2	3	-	2
PC16. identify the scope of improvement in the implementation of the mining plan and share appropriate feedback following the organizational procedure	2	4	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1425
NOS Name	Follow the mining plan and instructions for material handling
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Loading and Hauling - Opencast
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	11/03/2026
Next Review Date	11/03/2029
NSQC Clearance Date	11/03/2026

Qualification Pack

MIN/N1426: Carry out basic repair and maintenance of the mining shovel

Description

This unit is about carrying out basic repair and maintenance of the mining shovel.

Scope

The scope covers the following :

- Carry out cleaning
- Check and maintain the fluid levels
- Carryout lubrication and greasing
- Clean and replace the filters
- Clean the electrical system
- Maintain the hydraulic system
- Maintain the undercarriage
- Check the bucket or attachment
- Check and maintain the torque
- Inspect the safety devices
- Follow the manufacturer's guidelines

Elements and Performance Criteria

Carry out cleaning

To be competent, the user/individual on the job must be able to:

- PC1.** clean the exterior of the shovel, including the engine compartment, hydraulic system, and electrical components
- PC2.** remove dirt, dust, debris, and accumulated materials to prevent contamination and overheating
- PC3.** ensure the cleanliness of the operator's cabin, controls, windscreen, mirrors and other components

Check and maintain the fluid levels

To be competent, the user/individual on the job must be able to:

- PC4.** check the fluid levels including hydraulic fluid, engine oil, coolant, and fuel
- PC5.** refill the fluids as required to maintain the recommended fluid levels and ensure effective lubrication and cooling
- PC6.** inspect for signs of fluid leaks and take appropriate corrective measures to seal the leaks

Carry out lubrication and greasing

To be competent, the user/individual on the job must be able to:

- PC7.** carry out lubrication of the relevant parts and components, such as pivot points, joints, and bearings, following the manufacturer's recommendations
- PC8.** use the recommended grades of lubricant and grease for each application
- PC9.** apply grease to the grease points or fittings as specified in the shovel maintenance manual
- PC10.** follow the recommended greasing schedule to prevent premature wear

Qualification Pack

Clean and replace the filters

To be competent, the user/individual on the job must be able to:

- PC11.** clean or replace air filters, fuel filters, hydraulic filters, and other filters based on the manufacturer-recommended maintenance schedule
- PC12.** ensure filters are installed correctly as per the manufacturer's instructions

Check the electrical system

To be competent, the user/individual on the job must be able to:

- PC13.** examine the electrical wiring, components and connections for signs of corrosion, wear and tear or damage
- PC14.** test the shovel's lights, signals, and alarms for the correct functioning

Maintain the hydraulic system

To be competent, the user/individual on the job must be able to:

- PC15.** check the hydraulic hoses and fittings for leaks, wear, or damage, and replace the damaged components
- PC16.** examine the hydraulic system for appropriate pressure and flow
- PC17.** check for abnormal noises or vibrations in the hydraulic system and take the appropriate corrective measures

Maintain the undercarriage

To be competent, the user/individual on the job must be able to:

- PC18.** inspect the shovel tracks (as applicable) for wear, and appropriate tension and alignment
- PC19.** check the track rollers, idlers, and sprockets for damage or excessive wear

Check the bucket or attachment

To be competent, the user/individual on the job must be able to:

- PC20.** check the bucket or attachment for loose components, wear or damage
- PC21.** examine the shovel teeth, cutting edges, and wear plates to ensure they are in good condition

Check and maintain the torque

To be competent, the user/individual on the job must be able to:

- PC22.** check the torque settings on critical bolts and fasteners, including those on the bucket and attachment to ensure it is as recommended by the manufacturer
- PC23.** adjust the torque as per the requirement using the appropriate tools

Inspect the safety devices

To be competent, the user/individual on the job must be able to:

- PC24.** check the safety devices, such as seat belts, backup alarms, emergency shut-off switches, and fire extinguishers for the correct functioning
- PC25.** carry out appropriate repair and maintenance of the safety devices as required

Follow the manufacturer's guidelines

To be competent, the user/individual on the job must be able to:

- PC26.** follow the maintenance schedule, recommendations, and guidelines provided in the equipment manual for optimal equipment performance
- PC27.** use the manufacturer-recommended maintenance tools and spare parts

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of basic maintenance of a mining shovel to ensure the equipment's reliability, safety, and longevity, prevention of breakdowns, reduction in downtime, etc
- KU2.** the importance of carrying out maintenance of different components of the shovel as per the maintenance schedule recommended by the manufacturer
- KU3.** the importance of using the manufacturer-recommended maintenance tools and spare parts
- KU4.** different models of shovels used in mining operations
- KU5.** the components, systems, and technical specifications of different types of shovels
- KU6.** the safety protocols and procedures specific to maintenance of mining shovels
- KU7.** the potential hazards associated with mining shovel's maintenance and the use of appropriate Personal Protective Equipment (PPE)
- KU8.** how to read and follow the manufacturer's equipment maintenance manual for basic equipment maintenance
- KU9.** the recommended maintenance schedule, procedures, and checklists for a shovel's maintenance
- KU10.** the relevant tools, equipment, and materials required for a shovel's maintenance
- KU11.** the use of different hand and power tools, and specialized equipment
- KU12.** the types of fluids used in the shovel, i.e. hydraulic fluid, engine oil, coolant, and fuel
- KU13.** how to check fluid levels, assess fluid quality, and perform fluid changes
- KU14.** the recommended lubrication points on the shovel and the proper lubricants to be used
- KU15.** how to apply grease and lubricants to critical components to prevent wear and friction
- KU16.** the various filters, i.e. air filters, fuel filters, hydraulic filters, used in a shovel
- KU17.** how to replace or clean filters according to the maintenance schedule
- KU18.** the basic maintenance/troubleshooting of the shovel's electrical systems, including lights, alarms, sensors, and wiring
- KU19.** how to replace the damaged electrical components
- KU20.** the hydraulic systems, hoses, and fittings of the shovel
- KU21.** how to identify and address hydraulic leaks, replace hoses, and perform system checks
- KU22.** the importance of properly torqued bolts and fasteners in equipment safety and performance
- KU23.** the process of visually inspecting and checking the tightness of critical bolts and fasteners
- KU24.** the location and use of safety devices, such as emergency shut-off switches and fire suppression systems
- KU25.** how to test and maintain safety features
- KU26.** the environmental regulations and guidelines relevant to the maintenance of mining shovel
- KU27.** the importance and process of maintaining records of maintenance activities, including tasks performed, dates, and any issues identified
- KU28.** the utilization of maintenance logs and documentation to track equipment history
- KU29.** the emergency response procedures, including the location of emergency equipment and evacuation routes in case of accidents during maintenance

Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain the appropriate data and records
- GS2.** read the appropriate reports and literature concerning the field of work
- GS3.** communicate professionally with all the stakeholders
- GS4.** listen attentively to understand the information/ instructions being shared and take appropriate action
- GS5.** coordinate with co-workers to achieve the work objectives
- GS6.** plan tasks based on priority to ensure their timely completion
- GS7.** identify possible disruptions to work and take appropriate mitigation measures
- GS8.** take prompt action to deal with workplace emergencies and accidents
- GS9.** evaluate all possible solutions to work-related problems and select the best one
- GS10.** follow the recommended practices for the timely completion of work and achievement of organizational objectives

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out cleaning</i>	4	4	-	2
PC1. clean the exterior of the shovel, including the engine compartment, hydraulic system, and electrical components	2	2	-	1
PC2. remove dirt, dust, debris, and accumulated materials to prevent contamination and overheating	1	1	-	1
PC3. ensure the cleanliness of the operator's cabin, controls, windscreen, mirrors and other components	1	1	-	-
<i>Check and maintain the fluid levels</i>	3	5	-	3
PC4. check the fluid levels including hydraulic fluid, engine oil, coolant, and fuel	1	2	-	1
PC5. refill the fluids as required to maintain the recommended fluid levels and ensure effective lubrication and cooling	1	2	-	1
PC6. inspect for signs of fluid leaks and take appropriate corrective measures to seal the leaks	1	1	-	1
<i>Carry out lubrication and greasing</i>	4	8	-	4
PC7. carry out lubrication of the relevant parts and components, such as pivot points, joints, and bearings, following the manufacturer's recommendations	1	2	-	1
PC8. use the recommended grades of lubricant and grease for each application	1	2	-	1
PC9. apply grease to the grease points or fittings as specified in the shovel maintenance manual	1	2	-	1
PC10. follow the recommended greasing schedule to prevent premature wear	1	2	-	1
<i>Clean and replace the filters</i>	2	4	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. clean or replace air filters, fuel filters, hydraulic filters, and other filters based on the manufacturer-recommended maintenance schedule	1	2	-	1
PC12. ensure filters are installed correctly as per the manufacturer's instructions	1	2	-	-
<i>Check the electrical system</i>	3	4	-	1
PC13. examine the electrical wiring, components and connections for signs of corrosion, wear and tear or damage	1	2	-	1
PC14. test the shovel's lights, signals, and alarms for the correct functioning	2	2	-	-
<i>Maintain the hydraulic system</i>	3	5	-	3
PC15. check the hydraulic hoses and fittings for leaks, wear, or damage, and replace the damaged components	1	1	-	1
PC16. examine the hydraulic system for appropriate pressure and flow	1	2	-	1
PC17. check for abnormal noises or vibrations in the hydraulic system and take the appropriate corrective measures	1	2	-	1
<i>Maintain the undercarriage</i>	3	5	-	2
PC18. inspect the shovel tracks (as applicable) for wear, and appropriate tension and alignment	2	2	-	1
PC19. check the track rollers, idlers, and sprockets for damage or excessive wear	1	3	-	1
<i>Check the bucket or attachment</i>	2	3	-	1
PC20. check the bucket or attachment for loose components, wear or damage	1	2	-	1
PC21. examine the shovel teeth, cutting edges, and wear plates to ensure they are in good condition	1	1	-	-
<i>Check and maintain the torque</i>	2	4	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. check the torque settings on critical bolts and fasteners, including those on the bucket and attachment to ensure it is as recommended by the manufacturer	1	2	-	-
PC23. adjust the torque as per the requirement using the appropriate tools	1	2	-	1
<i>Inspect the safety devices</i>	2	4	-	1
PC24. check the safety devices, such as seat belts, backup alarms, emergency shut-off switches, and fire extinguishers for the correct functioning	1	2	-	1
PC25. carry out appropriate repair and maintenance of the safety devices as required	1	2	-	-
<i>Follow the manufacturer's guidelines</i>	2	4	-	1
PC26. follow the maintenance schedule, recommendations, and guidelines provided in the equipment manual for optimal equipment performance	1	2	-	1
PC27. use the manufacturer-recommended maintenance tools and spare parts	1	2	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1426
NOS Name	Carry out basic repair and maintenance of the mining shovel
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Loading and Hauling - Opencast
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	11/03/2026
Next Review Date	11/03/2029
NSQC Clearance Date	11/03/2026

Qualification Pack

MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines

Description

This unit is about adhering to health, safety and environmental guidelines at the Open Cast Mines while executing various tasks, maintenance and operations.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow Environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** comply with safety, health, and security-related regulations/guidelines at the opencast mine
- PC2.** follow the safety instructions given by the workman's inspector
- PC3.** follow adequate safety while working at haul roads, heights, overburden dumps, sump area, stockyard, near moving parts, etc.
- PC4.** take safety precautions while working on sites (sub-station, workshop etc.), with equipment, and conducting welding and cutting operations
- PC5.** follow appropriate Safe Operating Procedure (SOP) while dealing with explosives
- PC6.** respond promptly and appropriately to an accident/ incident or an emergency situation, within limits of the role and responsibility
- PC7.** provide first aid to an injured person
- PC8.** operate various types of fire extinguishers to control different types of fire at a worksite when required
- PC9.** use appropriate PPE as per the requirement
- PC10.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC11.** disinfect the machine/tools before and after work/task
- PC12.** maintain hygiene at the work site
- PC13.** report any symptoms of illness to the shift-incharge

Follow Environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC14.** identify the environmental impact of related opencast mining operations
- PC15.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC16.** ensure not to mix topsoil with waste in day to day tasks
- PC17.** ensure that HEMM is washed at the designated location

Qualification Pack

- PC18.** ensure the productivity of the machine for material/fuel conservation
- PC19.** follow the mineral conservation practices specified by the organization in accordance with MCDR-2017 (Mineral Conservation and Development Rules)
- PC20.** assist supervisor for reducing environmental impact caused due to related mining operations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** about various environmental awareness program related to mining, organized by the various government bodies/company
- KU2.** safety guidelines specified by Directorate General of Mine Safety (DGMS)
- KU3.** how to identify six-directional hazards
- KU4.** basic mining terminologies and definitions
- KU5.** importance of blind spot survey/training
- KU6.** means of access and egress from the mines, location of workshop, haul roads and working face including dump yards
- KU7.** duties of workers under The Mines act-1952
- KU8.** working hours and accident compensation as per The Mines act-1952
- KU9.** about precautions to be taken when handling heavy equipment
- KU10.** various problems/incidents likely to occur
- KU11.** hierarchy of the reporting
- KU12.** machine operation, condition of the machine and worksite
- KU13.** proper documents specific to the machine
- KU14.** role of workmen inspector, safety committee and internal safety organization
- KU15.** the process of top soil removal and management
- KU16.** mine sump and pumping system of the mines
- KU17.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU18.** common sources of pollution in the mines and ways to minimize it
- KU19.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, gas detector, safety lamp, self-contained breathing apparatus, gum boots, ear plugs, face mask, etc.
- KU20.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU21.** emergency response /disaster management plan prepared by the organization
- KU22.** signages, mining area-specific signs, and other safety and emergency signals
- KU23.** the outcome of violation of safety procedures
- KU24.** basic personal and workplace hygiene
- KU25.** importance of sensitization towards different genders and PWD (Persons with Disabilities)

Generic Skills (GS)

User/individual on the job needs to know how to:



Skill Council For Mining Sector



Qualification Pack

- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS4.** plan and organize the work order and jobs
- GS5.** use reasoning skills to identify and resolve fundamental problems
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	20	34	-	15
PC1. comply with safety, health, and security-related regulations/guidelines at the opencast mine	2	2	-	1
PC2. follow the safety instructions given by the workman's inspector	1	3	-	1
PC3. follow adequate safety while working at haul roads, heights, overburden dumps, sump area, stockyard, near moving parts, etc.	3	3	-	1
PC4. take safety precautions while working on sites (sub-station, workshop etc.), with equipment, and conducting welding and cutting operations	1	3	-	1
PC5. follow appropriate Safe Operating Procedure (SOP) while dealing with explosives	2	3	-	2
PC6. respond promptly and appropriately to an accident/ incident or an emergency situation, within limits of the role and responsibility	2	3	-	1
PC7. provide first aid to an injured person	2	3	-	1
PC8. operate various types of fire extinguishers to control different types of fire at a worksite when required	1	3	-	1
PC9. use appropriate PPE as per the requirement	2	4	-	2
PC10. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	1	2	-	1
PC11. disinfect the machine/tools before and after work/task	1	2	-	1
PC12. maintain hygiene at the work site	1	1	-	1
PC13. report any symptoms of illness to the shift-incharge	1	2	-	1
<i>Follow Environmental guidelines</i>	10	16	-	5

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. identify the environmental impact of related opencast mining operations	2	2	-	1
PC15. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	1	2	-	1
PC16. ensure not to mix topsoil with waste in day to day tasks	2	2	-	1
PC17. ensure that HEMM is washed at the designated location	2	2	-	1
PC18. ensure the productivity of the machine for material/fuel conservation	1	3	-	-
PC19. follow the mineral conservation practices specified by the organization in accordance with MCDR-2017 (Mineral Conservation and Development Rules)	1	3	-	1
PC20. assist supervisor for reducing environmental impact caused due to related mining operations	1	2	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1703
NOS Name	Follow Health, Safety, and Environmental Guidelines for opencast mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, , Exploration, Mineral Estimation, Planning, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Opencast, Specialized Operations, Electrical Services, Mechanical Services, Instrumentation and Control Systems, Ore Processing
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	11/03/2026
Next Review Date	11/03/2029
NSQF Clearance Date	11/03/2026

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

MIN/N1423: Operate and maintain the electric rope shovel

Description

This unit is about operating out maintaining the electric rope shovel.

Scope

The scope covers the following :

- Prepare for the electric rope shovel operations
- Operate the electric rope shovel
- Perform the post-operation activities

Elements and Performance Criteria

Prepare for the electric rope shovel operations

To be competent, the user/individual on the job must be able to:

- PC1.** evaluate the equipment and attachments for suitability
- PC2.** identify and address potential hazards and risks
- PC3.** select the appropriate Personal Protective Equipment (PPE)
- PC4.** carry out equipment pre-start checks
- PC5.** coordinate with the supervisor to position or move cable towers and power lines, as required
- PC6.** check the functioning of communication equipment to establish and maintain communication with other personnel

Operate the electric rope shovel

To be competent, the user/individual on the job must be able to:

- PC7.** follow the manufacturer's instructions for starting the electric rope shovel
- PC8.** start the electric rope shovel according to the manufacturer's operational requirements, warming up and exercising the shovel and associated equipment
- PC9.** check the controls and instruments for the correct functioning
- PC10.** use the operator room instruments and alarms as per the requirement
- PC11.** position the shovel on a level ground for safe and efficient extraction/ digging
- PC12.** operate the shovel following the manufacturer's instructions, with the correct reach angle to allow optimum efficiency and bucket load
- PC13.** adhere to the shovel's load capacity to minimize strain on the shovel and any consequent safety incidents
- PC14.** carry out digging and load trucks with the material
- PC15.** maintain the depth of bench to allow optimum use of bucket capacity
- PC16.** set and monitor bucket penetration for optimum bucket load and to minimize overload or strain on the shovel
- PC17.** walk the shovel and re-position the shovel, power lines and cable towers for safe and efficient operation
- PC18.** swing the shovel between the dig site and dumping location

Qualification Pack

PC19. dump the load into the trucks, and return the shovel to the excavation site

PC20. follow the appropriate procedures during equipment-related emergencies

PC21. ensure compliance with safety and environmental regulations

Perform the post-operation activities

To be competent, the user/individual on the job must be able to:

PC22. inspect the electric rope shovel and its attachments after use

PC23. identify and report any faults or defects

PC24. clean and maintain the electric rope shovel following the manufacturer's instructions

PC25. park and secure the electric rope shovel in the designated area

PC26. complete the relevant documentation and records concerning the use and inspection of the electric rope shovel

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the make and models of different electric rope shovels, including their technical specifications, controls, and features

KU2. the safety protocols and guidelines for operating the shovel, including knowledge of lockout/tagout procedures, personal protective equipment (PPE), and emergency response plans

KU3. the correct procedures for starting up and shutting down the electric rope shovel to ensure safe and efficient operation

KU4. the use of appropriate control systems in an electric rope shovel

KU5. the process of loading the shovel, including positioning, bucket filling, and dumping materials into haul trucks or conveyors

KU6. how to control the swing and hoist motions of the shovel's boom, including management of load swing and maintaining the balance during operation

KU7. different bucket types and sizes, and how to change and maintain them for specific tasks

KU8. efficient digging, including appropriate digging depth, angle, and cycle times, to maximize productivity

KU9. the handling of different types of materials, such as ore, overburden, or waste rock, and ensuring proper material flow and management

KU10. the manufacturer's maintenance guidelines, schedules, and recommendations for the electric rope shovel

KU11. the lubrication requirements for various components, including hoist drums, cables, bearings, and swing mechanisms

KU12. the maintenance of hydraulic systems, including pumps, cylinders, hoses, and filters, as applicable

KU13. the maintenance of electrical systems, control panels, sensors, and wiring used in the shovel's operation

KU14. the signs of wear, damage, or corrosion on critical components and appropriate maintenance

KU15. the importance of conducting regular inspections before and after use to identify any issues that may affect safety or performance

Qualification Pack

- KU16.** how to replace or repair worn or damaged components, such as cables, bearings, sensors, etc
- KU17.** the importance safety checks to be performed after maintenance to ensure that the shovel is safe for operation
- KU18.** the importance of maintaining detailed records of maintenance activities, including dates, tasks performed, parts replaced, and any issues identified
- KU19.** the emergency response procedures in case of accidents or unexpected equipment failures during operation and maintenance
- KU20.** the environmental regulations and best practices for handling fluids, fuels, and materials during maintenance to prevent contamination

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain the appropriate data and records
- GS2.** read the appropriate reports and literature concerning the field of work
- GS3.** communicate professionally with all the stakeholders
- GS4.** listen attentively to understand the information/ instructions being shared and take appropriate action
- GS5.** coordinate with co-workers to achieve the work objectives
- GS6.** plan tasks based on priority to ensure their timely completion
- GS7.** identify possible disruptions to work and take appropriate mitigation measures
- GS8.** take prompt action to deal with workplace emergencies and accidents
- GS9.** evaluate all possible solutions to work-related problems and select the best one
- GS10.** follow the recommended practices for the timely completion of work and achievement of organizational objectives

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for the electric rope shovel operations</i>	8	12	-	4
PC1. evaluate the equipment and attachments for suitability	2	3	-	1
PC2. identify and address potential hazards and risks	2	2	-	1
PC3. select the appropriate Personal Protective Equipment (PPE)	1	1	-	1
PC4. carry out equipment pre-start checks	1	2	-	1
PC5. coordinate with the supervisor to position or move cable towers and power lines, as required	1	2	-	-
PC6. check the functioning of communication equipment to establish and maintain communication with other personnel	1	2	-	-
<i>Operate the electric rope shovel</i>	16	28	-	13
PC7. follow the manufacturer's instructions for starting the electric rope shovel	2	2	-	1
PC8. start the electric rope shovel according to the manufacturer's operational requirements, warming up and exercising the shovel and associated equipment	1	2	-	1
PC9. check the controls and instruments for the correct functioning	1	2	-	1
PC10. use the operator room instruments and alarms as per the requirement	1	2	-	1
PC11. position the shovel on a level ground for safe and efficient extraction/ digging	1	2	-	1
PC12. operate the shovel following the manufacturer's instructions, with the correct reach angle to allow optimum efficiency and bucket load	1	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. adhere to the shovel's load capacity to minimize strain on the shovel and any consequent safety incidents	1	2	-	1
PC14. carry out digging and load trucks with the material	1	2	-	1
PC15. maintain the depth of bench to allow optimum use of bucket capacity	1	2	-	1
PC16. set and monitor bucket penetration for optimum bucket load and to minimize overload or strain on the shovel	1	2	-	1
PC17. walk the shovel and re-position the shovel, power lines and cable towers for safe and efficient operation	1	2	-	1
PC18. swing the shovel between the dig site and dumping location	1	2	-	1
PC19. dump the load into the trucks, and return the shovel to the excavation site	1	2	-	1
PC20. follow the appropriate procedures during equipment-related emergencies	1	1	-	-
PC21. ensure compliance with safety and environmental regulations	1	1	-	-
<i>Perform the post-operation activities</i>	6	10	-	3
PC22. inspect the electric rope shovel and its attachments after use	2	2	-	1
PC23. identify and report any faults or defects	1	2	-	1
PC24. clean and maintain the electric rope shovel following the manufacturer's instructions	1	2	-	1
PC25. park and secure the electric rope shovel in the designated area	1	2	-	-
PC26. complete the relevant documentation and records concerning the use and inspection of the electric rope shovel	1	2	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1423
NOS Name	Operate and maintain the electric rope shovel
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Loading and Hauling - Opencast
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	11/03/2026
Next Review Date	11/03/2029
NSQC Clearance Date	11/03/2026

Qualification Pack

MIN/N1424: Operate and maintain the hydraulic shovel

Description

This unit is about operating out maintaining the hydraulic shovel

Scope

The scope covers the following :

- Perform pre-operation checks on the hydraulic shovel
- Operate the hydraulic shovel
- Maintain communication and safety

Elements and Performance Criteria

Perform pre-operation checks on the hydraulic shovel

To be competent, the user/individual on the job must be able to:

- PC1.** perform the pre-operation inspection of the hydraulic shovel, checking for visible damage, loose or missing parts, and tire or track condition
- PC2.** inspect hydraulic hoses and connections for leaks or damage and take appropriate corrective measures
- PC3.** repair or replace any worn or damaged components promptly
- PC4.** ensure the safety devices and controls are functioning correctly

Operate the hydraulic shovel

To be competent, the user/individual on the job must be able to:

- PC5.** start the hydraulic shovel following the manufacturer's instructions, allowing the engine to warm up for the appropriate duration
- PC6.** ensure all controls are in neutral before starting the engine
- PC7.** use the shovel controls as per the manufacturer's instructions
- PC8.** position the shovel's bucket or attachment safely for excavation and loading operations
- PC9.** carry out excavation and loading of material safely
- PC10.** lift the material appropriately, lowering the bucket into the material, closing it, and lifting it to the required height
- PC11.** use the appropriate loading techniques to prevent the overloading of shovel and unstable loads
- PC12.** deposit the material into a haul truck or designated area, swinging the shovel steadily, avoiding excessive swinging or sudden movements
- PC13.** engage the hydraulic shovel's bucket or attachment and raise it to a safe height for travel
- PC14.** follow the appropriate procedure for shutting down the shovel
- PC15.** park the shovel at the designated location, lowering the bucket or attachment to the ground and engaging the parking brake

Maintain communication and safety

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC16.** maintain clear communication with ground personnel and other equipment operators to ensure safe operations
- PC17.** check the availability of appropriate Personal Protective Equipment (PPE), e.g. hard hat, safety glasses, high-visibility vest, steel-toed boots, etc
- PC18.** use the required PPE at the mining site and during equipment operation
- PC19.** follow all the site safety procedures and protocols established by the organization
- PC20.** use appropriate hand signals or radios as required
- PC21.** maintain awareness of the surroundings to avoid any accidents
- PC22.** follow the applicable safety rules, such as appropriate speed limits for the shovel and ensure no personnel or obstacles in the work area before moving or swinging the shovel

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the make and models of different hydraulic shovels, including their technical specifications, controls, and features
- KU2.** the safety protocols and guidelines for operating the shovel, including knowledge of lockout/tagout procedures, personal protective equipment (PPE), and emergency response plans
- KU3.** the correct procedures for starting up and shutting down the hydraulic shovel to ensure safe and efficient operation
- KU4.** the use of appropriate control systems in an hydraulic shovel
- KU5.** the process of loading the shovel, including positioning, bucket filling, and dumping materials into haul trucks or conveyors
- KU6.** how to control the swing and hoist motions of the shovel's boom, including management of load swing and maintaining the balance during operation
- KU7.** different bucket types and sizes, and how to change and maintain them for specific tasks
- KU8.** efficient digging, including appropriate digging depth, angle, and cycle times, to maximize productivity
- KU9.** the handling of different types of materials, such as ore, overburden, or waste rock, and ensuring proper material flow and management
- KU10.** the manufacturer's maintenance guidelines, schedules, and recommendations for the hydraulic shovel
- KU11.** the lubrication requirements for various components, including hoist drums, cables, bearings, and swing mechanisms
- KU12.** the maintenance of hydraulic systems, including pumps, cylinders, hoses, and filters, as applicable
- KU13.** the maintenance of electrical systems, control panels, sensors, and wiring used in the shovel's operation
- KU14.** the signs of wear, damage, or corrosion on critical components and appropriate maintenance
- KU15.** the importance of conducting regular inspections before and after use to identify any issues that may affect safety or performance

Qualification Pack

- KU16.** how to replace or repair worn or damaged components, such as cables, bearings, sensors, etc.
- KU17.** the importance safety checks to be performed after maintenance to ensure that the shovel is safe for operation
- KU18.** the importance of maintaining detailed records of maintenance activities, including dates, tasks performed, parts replaced, and any issues identified
- KU19.** the emergency response procedures in case of accidents or unexpected equipment failures during operation and maintenance
- KU20.** the environmental regulations and best practices for handling fluids, fuels, and materials during maintenance to prevent contamination

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain the appropriate data and records
- GS2.** read the appropriate reports and literature concerning the field of work
- GS3.** communicate professionally with all the stakeholders
- GS4.** listen attentively to understand the information/ instructions being shared and take appropriate action
- GS5.** coordinate with co-workers to achieve the work objectives
- GS6.** plan tasks based on priority to ensure their timely completion
- GS7.** identify possible disruptions to work and take appropriate mitigation measures
- GS8.** take prompt action to deal with workplace emergencies and accidents
- GS9.** evaluate all possible solutions to work-related problems and select the best one
- GS10.** follow the recommended practices for the timely completion of work and achievement of organizational objectives

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform pre-operation checks on the hydraulic shovel</i>	8	8	-	4
PC1. perform the pre-operation inspection of the hydraulic shovel, checking for visible damage, loose or missing parts, and tire or track condition	2	2	-	1
PC2. inspect hydraulic hoses and connections for leaks or damage and take appropriate corrective measures	2	2	-	1
PC3. repair or replace any worn or damaged components promptly	2	2	-	1
PC4. ensure the safety devices and controls are functioning correctly	2	2	-	1
<i>Operate the hydraulic shovel</i>	13	28	-	10
PC5. start the hydraulic shovel following the manufacturer's instructions, allowing the engine to warm up for the appropriate duration	2	2	-	1
PC6. ensure all controls are in neutral before starting the engine	2	2	-	1
PC7. use the shovel controls as per the manufacturer's instructions	1	2	-	1
PC8. position the shovel's bucket or attachment safely for excavation and loading operations	1	3	-	1
PC9. carry out excavation and loading of material safely	1	2	-	1
PC10. lift the material appropriately, lowering the bucket into the material, closing it, and lifting it to the required height	1	3	-	1
PC11. use the appropriate loading techniques to prevent the overloading of shovel and unstable loads	1	3	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. deposit the material into a haul truck or designated area, swinging the shovel steadily, avoiding excessive swinging or sudden movements	1	3	-	1
PC13. engage the hydraulic shovel's bucket or attachment and raise it to a safe height for travel	1	3	-	1
PC14. follow the appropriate procedure for shutting down the shovel	1	3	-	1
PC15. park the shovel at the designated location, lowering the bucket or attachment to the ground and engaging the parking brake	1	2	-	-
<i>Maintain communication and safety</i>	9	14	-	6
PC16. maintain clear communication with ground personnel and other equipment operators to ensure safe operations	2	2	-	1
PC17. check the availability of appropriate Personal Protective Equipment (PPE), e.g. hard hat, safety glasses, high-visibility vest, steel-toed boots, etc	2	2	-	1
PC18. use the required PPE at the mining site and during equipment operation	1	2	-	1
PC19. follow all the site safety procedures and protocols established by the organization	1	2	-	1
PC20. use appropriate hand signals or radios as required	1	2	-	1
PC21. maintain awareness of the surroundings to avoid any accidents	1	2	-	1
PC22. follow the applicable safety rules, such as appropriate speed limits for the shovel and ensure no personnel or obstacles in the work area before moving or swinging the shovel	1	2	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1424
NOS Name	Operate and maintain the hydraulic shovel
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Loading and Hauling - Opencast
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	11/03/2026
Next Review Date	11/03/2029
NSQF Clearance Date	11/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1419.Perform the pre-operation checks on the shovel	30	50	0	20	100	10
MIN/N1420.Operate the shovel for mining operations	30	50	0	20	100	20
MIN/N1421.Perform the post-shovel operation activities	30	50	0	20	100	15
MIN/N1422.Carry out appropriate activities before and after the shovel maintenance	30	50	0	20	100	10
MIN/N1425.Follow the mining plan and instructions for material handling	30	50	0	20	100	5
MIN/N1426.Carry out basic repair and maintenance of the mining shovel	30	50	0	20	100	5
MIN/N1703.Follow Health, Safety, and Environmental Guidelines for opencast mines	30	50	-	20	100	10
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5
Total	230	380	-	140	750	80

Elective: 1 Electric Rope Shovel



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1423. Operate and maintain the electric rope shovel	30	50	0	20	100	20
Total	30	50	-	20	100	20

Elective: 2 Hydraulic Shovel

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1424. Operate and maintain the hydraulic shovel	30	50	0	20	100	20
Total	30	50	-	20	100	20

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
QP	Qualification Pack
NSQF	National Skills Qualifications Framework
TVET	Technical and Vocational Education and Training
SCMS	Skill Council for Mining Sector
ROPS	Roll-Over Protection Structure
FOPS	Falling Object Protective Structure
PC	Performance Criteria
HEMM	Heavy Earth Moving Machinery
OEM	Original Equipment Manufacturer
DGMS	Directorate General of Mining Safety
MMR	Metalliferous Mines Regulations
CMR	Coal Mines Regulations
SOP	Standard Operating Practice
MVTR	Mine Vocational Training Rules
MRR	Mines Rescue Rules
HSE	health, safety & environment
OB	Over Burden
ERP	Enterprise resource planning
MCDR	Mineral Conservation and Development Rules

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	A sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	a sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in the industry.
Occupational Standards	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standard	NOS are occupational standards which apply uniquely in the Indian context.

Qualification Pack

Qualification Pack	QP comprises the set of OS, together with the educational, training, and other criteria required to perform a job role. A QP is assigned a unique qualifications pack role
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.